

Lecture 5: Constrained Optimization II

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This lecture focuses on three other aspects of the Lagrangian approach:

- ▶ the sensitivity of the optimal value of the objective function to changes in the parameters
- ▶ the second order conditions that distinguish maxima from minima
- ▶ the constraint qualifications that are a subtle but necessary hypothesis in the Lagrangian approach

- 1 The Meaning of the Multiplier
- 2 Envelope Theorems
- 3 Second Order Conditions
- 4 Smooth Dependence on the Parameters
- 5 Constraint Qualifications

Consider the simplest problem with one equality constraint:

$$\max f(x, y) \quad (1)$$

$$\text{s.t. } h(x, y) = a \quad (2)$$

Theorem 19.1

- ▶ Let f and h be C^1 functions of two variables. For any fixed value of the parameter a , let $(x^*(a), y^*(a))$ be the solution of problem (1)–(2) with corresponding multiplier $\mu^*(a)$.
- ▶ Suppose that x^*, y^* and μ^* are C^1 functions of a and that NDCQ holds at $(x^*(a), y^*(a))$. Then

$$\mu^*(a) = \frac{d}{da} f(x^*(a), y^*(a))$$

Example 19.1: Previously, we found that a maximizer of $f(x_1, x_2) = x_1^2 x_2$ on the constraint set $2x_1^2 + x_2^2 = 3$ is $x_1 = x_2 = 1$, with multiplier $\mu = 0.5$.

- ▶ The maximum value of f is $f^* = f(1, 1) = 1$
- ▶ Redo the problem, this time using the constraint

$$2x_1^2 + x_2^2 = 3.3$$

- ▶ The same computation yields the solution $x_1 = x_2 = \sqrt{1.1}$, with maximum value $f^* = (1.1)^{3/2} \approx 1.1537$, an increase of 0.1537 over the original f^*

Example 19.1:

- ▶ On the other hand, Theorem 19.1 predicts that changing the right-hand side of the constraint by 0.3 unit would change the maximum value of the objective function by roughly

$$0.3\mu = 0.3 * 0.5 = 0.15 \text{ unit,}$$

an approximation correct to two decimal places

Theorem 19.2

Let $f, h_1, \dots, h_m$ be C^1 functions on $\mathbb{R}^n$. Let $\mathbf{a} = (a_1, \dots, a_m)$ be an m -tuple of exogenous parameters, and consider the problem of maximizing $f(x_1, \dots, x_n)$ subject to

$$h_1(x_1, \dots, x_n) = a_1, \dots, h_m(x_1, \dots, x_n) = a_m.$$

Let $x_1^*(\mathbf{a}), \dots, x_n^*(\mathbf{a})$ denote the solution of the problem with corresponding Lagrange multipliers $\mu_1^*(\mathbf{a}), \dots, \mu_m^*(\mathbf{a})$. Suppose further that the x_i^* 's and μ_j^* 's are differentiable functions of $(a_1, \dots, a_m)$ and that NDCQ holds. Then, for each $j = 1, \dots, m$,

$$\mu_j^*(a_1, \dots, a_m) = \frac{\partial}{\partial a_j} f(x_1^*(a_1, \dots, a_m), \dots, x_n^*(a_1, \dots, a_m))$$

Theorem 19.3

Let $\mathbf{a}^* = (a_1^*, \dots, a_k^*)$ be a k -tuple. Consider the problem of maximizing $f(x_1, \dots, x_n)$ subject to

$$g_1(x_1, \dots, x_n) \leq a_1^*, \dots, g_k(x_1, \dots, x_n) \leq a_k^*.$$

Let $x_1^*(\mathbf{a}^*), \dots, x_n^*(\mathbf{a}^*)$ denote the solution of the problem with corresponding Lagrange multipliers $\lambda_1^*(\mathbf{a}^*), \dots, \lambda_k^*(\mathbf{a}^*)$. Suppose that as $\mathbf{a}$ varies near $\mathbf{a}^*$, $x_1^*, \dots, x_n^*$ and $\lambda_1^*, \dots, \lambda_k^*$ are differentiable functions of $(a_1, \dots, a_k)$ and that NDCQ holds at $\mathbf{a}^*$. Then, for each $j = 1, \dots, k$,

$$\lambda_j^*(a_1^*, \dots, a_k^*) = \frac{\partial}{\partial a_j} f(x_1^*(a_1^*, \dots, a_k^*), \dots, x_n^*(a_1^*, \dots, a_k^*))$$

Think of the objective function as the profit function of a firm and think of a_j as the amount of available input in the production process. Then $\lambda_j = \partial f / \partial a_j$ is called the *internal value*, or *shadow price* of input j :

- ▶ It tells how valuable another unit of certain input would be to the firm's profits
- ▶ It tells the maximum amount the firm would be willing to pay to acquire another unit of the input

Example 19.2 Previously, we computed that the maximizer of xyz on the constraint set

$$x + y + z \leq 1, x \geq 0, y \geq 0, z \geq 0$$

is $x = y = z = 1/3$, where $xyz = 1/27$. The four multipliers are $1/9, 0, 0$ and 0 respectively.

- (a) If we change the first constraint to $x + y + z \leq 0.9$, we compute that the solution occurs at $x = y = z = 0.3$, where $xyz = 0.027$. Theorem 19.3 predicts that the new optimal value would be $\frac{1}{27} + \frac{1}{9} \cdot \left(-\frac{1}{10}\right) \approx 0.0259$, an estimate that is off by only 0.0011 or 4%.
- (b) If, instead, we change the second constraint from $x \geq 0$ to $x \geq 0.1$, we do not change the solution or the optimum value because the new region is a subset of the old region and it still contains the optimal point for the old region. This result is consistent with Theorem 19.3 since the multiplier for the (nonbinding) constraint $x \geq 0$ was zero.

- 1 The Meaning of the Multiplier
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- ▶ Previous theorems are special cases of a class of theorems which describe how the optimal value of the objective function in a parameterized optimization problem changes as one of the parameter changes.
- ▶ Such theorems are called **Envelope theorems**.

Theorem 19.4

Let $f(\mathbf{x}; a)$ be a C^1 function of $\mathbf{x} \in \mathbb{R}^n$ and the scalar a . For each choice of the parameter a , consider the unconstrained maximization problem: $\max_{\mathbf{x}} f(\mathbf{x}; a)$. Let $\mathbf{x}^*(a)$ be a solution of this problem. Suppose that $\mathbf{x}^*(a)$ is a C^1 function of a . Then

$$\frac{d}{da} f(\mathbf{x}^*(a); a) = \frac{\partial}{\partial a} f(\mathbf{x}^*(a); a)$$

Example 19.3 Consider the problem

$$\max_x f(x, a) = -a^3x^4 + 15x^3 - e^ax^2 + 17$$

around $a = 1$. Since f is a quartic polynomial in x with a negative leading coefficient, when $a = 1$, $f \rightarrow -\infty$ as $x \rightarrow \pm\infty$. So f admits a finite global maximizer $x^*(a)$ for each value of a near 1. Hence,

$$\frac{d}{da}f(x^*(a), a) = \frac{\partial}{\partial a}f(x^*(a), a) = -3a^2x^{*4} - e^ax^{*2} < 0$$

Without solving for the optimal $x^*(a)$, we can tell that as a increases beyond 1, $f(x^*(a), a)$ is a decreasing function of a . The peak of the graph of the function decreases as a increases.

Example 19.4 What will be the effect of a unit increase in a on the value of

$$\max_x f(x, a) = -x^2 + 2ax + 4a^2$$

for each a ? Since $f'(x) = -2x + 2a = 0$, $x^*(a) = a$. Then

$$f(x^*(a), a) = f(a, a) = -a^2 + 2a * a + 4a^2 = 5a^2,$$

which will increase at a rate of $10a$ as a increases. Instead, if we apply the envelope theorem,

$$\frac{df^*}{da} = \frac{\partial f}{\partial a}(x^*(a), a) = 2x + 8a = 10a$$

since $x^*(a) = a$.

Example 19.5

- ▶ A silicon Valley firm produces an output of microchips denoted by y and has a cost function $c(y)$ with $c'(y) > 0$ and $c''(y) > 0$.
- ▶ Of the chips it produces, a fraction $1 - \alpha$ are unavoidably defective and cannot be sold.
- ▶ Working chips can be sold at price p , and the microchip market is highly competitive.

How will an increase in production quality affect the firm's profit?

Example 19.5 The firm's profit function is given by

$$\pi(p, \alpha) = \max_y [p\alpha y - c(y)]$$

- ▶ The conditions on the cost function guarantee that there is a nonzero profit-maximizing output $y^*(\alpha)$ which depends smoothly on α .
- ▶ The derivative of optimal profit π with respect to α is

$$\frac{d\pi}{d\alpha} = \frac{\partial}{\partial \alpha}(p\alpha y - c(y)) = py > 0$$

Theorem 19.5

Let $f, h_1, \dots, h_k: \mathbb{R}^n \times \mathbb{R}^1 \rightarrow \mathbb{R}^1$ be C^1 functions. Let $\mathbf{x}^*(a) = (x_1^*(a), \dots, x_n^*(a))$ denote the solution of the problem of maximizing $\mathbf{x} \rightarrow f(\mathbf{x}, a)$ on the constraint set

$$h_1(\mathbf{x}; a) = 0, \dots, h_k(\mathbf{x}; a) = 0$$

for any fixed choice of the parameter a . Suppose that $\mathbf{x}^*(a)$ and the Lagrangian multipliers $\mu_1(a), \dots, \mu_k(a)$ are C^1 functions of a and that the NDCQ holds. Then,

$$\frac{d}{da} f(\mathbf{x}^*(a); a) = \frac{\partial L}{\partial a}(\mathbf{x}^*(a), \mu(a); a).$$

Example 19.6 Consider the problem

$$\begin{aligned} \max \quad & f(x, y) = xy \\ \text{s.t.} \quad & x^2 + y^2 \leq 1 \end{aligned}$$

Now change the constraint to $x^2 + 1.1y^2 \leq 1$. If we write the constraint as $x^2 + ay^2 \leq 1$, the Lagrangian for the parameterized problem is $L(x, y, \lambda; a) = xy - \lambda(x^2 + ay^2 - 1)$.

- ▶ The solution for the original ($a = 1$) problem was $x = y = \frac{1}{\sqrt{2}}$, $\lambda = \frac{1}{2}$
- ▶ The Envelope Theorem tells us that as a changes from 1 to 1.1, the optimal value of f changes by approximately $\frac{\partial L}{\partial a} \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{2}; 1 \right) * 0.1$
- ▶ Since $\frac{\partial L}{\partial a} = -\lambda y^2 = -1/4$, the optimal value will decrease by approximately $0.1 * (1/4) = 0.025$ to 0.475
- ▶ Calculate directly that the solution to the new problem is $x = \frac{1}{\sqrt{2}}, y = \frac{1}{\sqrt{2.2}}$, with $f^* \approx 0.4767$.

- 1 The Meaning of the Multiplier
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Theorem 19.7

Let f and h be C^2 functions on $\mathbb{R}^2$. Consider the problem of maximizing f on the constraint set $C_h = \{(x, y) : h(x, y) = c\}$. Form the Lagrangian function

$$L(x, y, \mu) = f(x, y) - \mu[h(x, y) - c].$$

Suppose that (x^*, y^*, μ^*) satisfies

$$\triangleright \frac{\partial L}{\partial x} = 0, \frac{\partial L}{\partial y} = 0, \frac{\partial L}{\partial \mu} = 0 \text{ at } (x^*, y^*, \mu^*)$$

$$\triangleright \det \begin{pmatrix} 0 & \frac{\partial h}{\partial x} & \frac{\partial h}{\partial y} \\ \frac{\partial h}{\partial x} & \frac{\partial^2 L}{\partial x^2} & \frac{\partial^2 L}{\partial x \partial y} \\ \frac{\partial h}{\partial y} & \frac{\partial^2 L}{\partial x \partial y} & \frac{\partial^2 L}{\partial y^2} \end{pmatrix} > 0$$

Then (x^*, y^*) is a local maximizer of f .

Theorem 19.6

Let f and $h_1, \dots, h_k$ be C^2 functions on $\mathbb{R}^n$. Consider the problem of maximizing f on the constraint set

$$C_h \equiv \{\mathbf{x} : h_1(\mathbf{x}) = c_1, \dots, h_k(\mathbf{x}) = c_k\}$$

Form the Lagrangian and suppose that:

- ▶ $\mathbf{x}^*$ lies in the constraint set C_h ;
- ▶ There exist $\mu_1^*, \dots, \mu_k^*$ such that

$$\frac{\partial L}{\partial x_1} = 0, \dots, \frac{\partial L}{\partial x_n} = 0, \frac{\partial L}{\partial \mu_1} = 0, \dots, \frac{\partial L}{\partial \mu_k} = 0$$

at $(x_1^*, \dots, x_n^*, \mu_1^*, \dots, \mu_k^*)$

- ▶ the Hessian of L with respect to $\mathbf{x}$ at $(\mathbf{x}^*, \mu^*)$ is negative definite on the linear constraint set $\{\mathbf{v} : D\mathbf{h}(\mathbf{x}^*)\mathbf{v} = 0\}$

Then $\mathbf{x}^*$ is a strict local constrained max of f on C_h .

Example 19.7 In example 18.5, we consider the problem:

$$\begin{aligned} &\text{maximize} && f(x_1, x_2) = x_1^2 x_2 \\ &\text{s.t.} && C_h = \{(x_1, x_2) : 2x_1^2 + x_2^2 = 3\}. \end{aligned}$$

We found six solutions to the first order conditions:

$$(x_1, x_2, \mu) = \begin{cases} (0, \pm\sqrt{3}, 0) \\ (\pm 1, +1, +0.5) \\ (\pm 1, -1, -0.5) \end{cases}$$

Example 19.7 Let's use the second order conditions to decide which of these points are local maxima and which are local minima. The Hessian is given by

$$H = \begin{pmatrix} 0 & h_{x_1} & h_{x_2} \\ h_{x_1} & L_{x_1 x_1} & L_{x_1 x_2} \\ h_{x_2} & L_{x_2 x_1} & L_{x_2 x_2} \end{pmatrix} = \begin{pmatrix} 0 & 4x_1 & 2x_2 \\ 4x_1 & 2x_2 - 4\mu & 2x_1 \\ 2x_2 & 2x_1 & -2\mu \end{pmatrix}$$

This problem has $n = 2$ variables and $k = 1$ equality constraints.

Example 19.7 As Theorem 19.7 indicates, we need only check the sign of $n - k$ determinant—the determinant of H itself.

- At the points $(\pm 1, -1, -0.5)$:

$$H = \begin{pmatrix} 0 & \pm 4 & -2 \\ \pm 4 & 0 & \pm 2 \\ -2 & \pm 2 & 1 \end{pmatrix}$$

In either case, $\det H = -48$; so these two points are local minima.

- At the points $(\pm 1, 1, 0.5)$:

$$H = \begin{pmatrix} 0 & \pm 4 & 2 \\ \pm 4 & 0 & \pm 2 \\ 2 & \pm 2 & -1 \end{pmatrix}$$

In either case, $\det H = 48$; so these two points are local maxima.

Example 19.7

- At the points $(0, \pm\sqrt{3}, 0)$, the corresponding bordered Hessian is

$$H = \begin{pmatrix} 0 & 0 & \pm 2\sqrt{3} \\ 0 & \pm 2\sqrt{3} & 0 \\ \pm 2\sqrt{3} & 0 & 0 \end{pmatrix}$$

For $(x_1, x_2) = (0, +\sqrt{3})$, $\det H = -24\sqrt{3} < 0$, this point is a local min.

For $(x_1, x_2) = (0, -\sqrt{3})$, $\det H = +24\sqrt{3} > 0$, this point is a local max.

Example 19.8 Consider the problem:

$$\begin{aligned} \max \quad & f(x, y, z) = x^2 y^2 z^2 \\ \text{s.t.} \quad & C_h = \{(x, y, z) : x^2 + y^2 + z^2 = 3\}. \end{aligned}$$

The first order conditions are:

$$\begin{aligned} \partial L / \partial x &= 2xy^2z^2 - 2\mu x = 0 \\ \partial L / \partial y &= 2x^2yz^2 - 2\mu y = 0 \\ \partial L / \partial z &= 2x^2y^2z - 2\mu z = 0 \\ -\partial L / \partial \mu &= x^2 + y^2 + z^2 - 3 = 0 \end{aligned}$$

with solution $x^2 = y^2 = z^2 = \mu = 1$.

Example 19.8 The bordered Hessian for this problem is:

$$H = \begin{pmatrix} 0 & 2x & 2y & 2z \\ 2x & 2y^2z^2 - 2\mu & 4xyz^2 & 4xy^2z \\ 2y & 4xyz^2 & 2x^2z^2 - 2\mu & 4x^2yz \\ 2z & 4xy^2z & 4x^2yz & 2x^2y^2 - 2\mu \end{pmatrix}$$

Example 19.8 At $x = y = z = \mu = 1$, the bordered Hessian becomes

$$H = \left(\begin{array}{ccc|c} 0 & 2 & 2 & 2 \\ 2 & 0 & 4 & 4 \\ 2 & 4 & 0 & 4 \\ \hline - & - & - & - \\ 2 & 4 & 4 & 0 \end{array} \right)$$

Since $n = 3$ and $k = 1$, we have to check the signs of the two leading principal minors: $\det H_3 = 32$ and $\det H_4 = -192$. Hence, the candidate $x = y = z = 1$ is local constrained max by Theorem 19.6.

Theorem 19.8 (Mixed Constraints)

Let $f, g_1, \dots, g_m$ and $h_1, \dots, h_k$ be C^2 functions on $\mathbb{R}^n$. Consider the problem of maximizing f on

$$C_{g,h} \equiv \{\mathbf{x} : g_1(\mathbf{x}) \leq b_1, \dots, g_m(\mathbf{x}) \leq b_m, h_1(\mathbf{x}) = c_1, \dots, h_k(\mathbf{x}) = c_k\}.$$

Form the Lagrangian

$$\begin{aligned} L(\mathbf{x}, \lambda_1, \dots, \lambda_m, \mu_1, \dots, \mu_k) = & f(\mathbf{x}) - \lambda_1[g_1(\mathbf{x}) - b_1] - \dots - \lambda_m[g_m(\mathbf{x}) - b_m] \\ & - \mu_1[h_1(\mathbf{x}) - c_1] - \dots - \mu_k[h_k(\mathbf{x}) - c_k] \end{aligned}$$

- ▶ Suppose that there exist $\lambda_1^*, \dots, \lambda_m^*, \mu_1^*, \dots, \mu_k^*$ such that the first order conditions are satisfied.
- ▶ Suppose only $g_1, \dots, g_e$ are binding at $\mathbf{x}^*$. Write $(g_1, \dots, g_e)$ as g_E . Suppose that the Hessian of L with respect to $\mathbf{x}$ at $(\mathbf{x}^*, \lambda^*, \mu^*)$ is negative definite on the linear constraint set $\{\mathbf{v} : Dg_E(\mathbf{x}^*)\mathbf{v} = 0 \text{ and } Dh(\mathbf{x}^*)\mathbf{v} = 0\}$.

Then $\mathbf{x}^*$ is a strict local constrained max of f on $C_{g,h}$.

- 1 The Meaning of the Multiplier
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Theorem 19.9

Consider the problem of maximizing $f(x; a)$ subject to $h_1(x; a) = 0$, $h_2(x; a) = 0$, $\dots$, $h_k(x; a) = 0$. Let $x^*(a)$ be the solution of the parameterized constrained maximization problem and let $\mu^*(a)$ be the corresponding Lagrange multiplier. If the Hessian matrix of the Lagrangian is nonsingular at the point $(x^*(a_0), \mu^*(a_0); a_0)$, then:

- 1 $x^*(a)$ and $\mu^*(a)$ are C^1 functions of a at $a = a_0$; and
- 2 the NDCQ holds at $(x^*(a_0), \mu^*(a_0); a_0)$.

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- ▶ In Lecture 4, we talked about NDCQ
- ▶ A theorem that does not impose constraint qualification: Theorem 19.11
- ▶ There are other constraint qualifications
- ▶ Theorem 19.12